Unforgeable Process Evidence: A New Paradigm of Scientific Integrity Complementing Reproducibility

Chong Liu*

August 15, 2026

Abstract

Reproducibility has become the dominant frame for scientific integrity: a result is trusted insofar as an independent party can rebuild it from shared data and code. We argue that reproducibility verifies *what* was claimed but remains silent about *how* the claim was arrived at, and that this second question is a distinct and complementary axis of integrity. We develop that axis around a single graded concept, *unforgeability*: a property of *evidence*, not of any particular record, that measures how hard it would be to produce a convincing counterfeit. We argue that the new paradigm this concept underwrites is not a tool but a *system of community review and knowledge-establishment*, in which confidence in a scientific claim can rest on the unforgeability of its evidence chain and on the soundness of the reasoning from evidence to claim, rather than on reproducibility alone—much as confidence in a mathematical theorem rests on the checkability of its proof chain and the soundness of inference, not on rerunning it. We are careful about what this does *not* establish: unforgeability secures the evidence, not the world it describes; it does not by itself show that the underlying events occurred, nor truth, originality, authorship, or completeness. Its epistemic role is therefore *defeater reduction*: a validated record eliminates a specified class of provenance doubts—post-hoc modification, backdating, silent deletion or reordering—rather than certifying the conduct itself. We show that the property is attainable with local, privacy-preserving, cryptographic recording that any third party can verify offline, and we instantiate it in a proof-of-concept exemplar: an open specification (Evidence Package v1) and a prototype reference implementation (Academic Integrity Recorder), released as open source. We close with scope, limitations, and a governance path for adoption.

Keywords: unforgeability; reproducibility; research integrity; process provenance; evidence.

1 Introduction

The reproducibility crisis is usually told as a story about *results*: too many published findings cannot be rebuilt from the materials that authors share [Nosek et al., 2015, Donoho et al., 2009]. The standard remedy has therefore been to share more—data, code, protocols, preregistrations—so that an independent party can re-derive the outcome [Nosek et al., 2015, Donoho et al., 2009]. This framing has made *reproducibility* the de facto currency of scientific integrity: a result is trustworthy to the extent that it can be regenerated.

*Corresponding author. Independent Researcher. ORCID: 0009-0009-8116-1255. Email: chong.liu.phil@outlook.com. Code and specification: `https://github.com/chongliuresearch/academic-integrity-recorder`.

We propose a complementary frame. Reproducibility answers the question *"can the result be rebuilt?"* It does not answer *"was the result arrived at honestly?"* A finding can be perfectly reproducible and yet be the product of selective reporting, post-hoc hypothesizing, undisclosed reuse of a collaborator's work, or an AI-assisted draft presented as original. Conversely, a genuinely honest workflow may fail to reproduce for reasons unrelated to misconduct—fragile materials, unreported environmental conditions, or legitimate stochasticity [Ioannidis, 2005]. The two questions are correlated but distinct, and the crisis literature documents the blind spot explicitly: even when a study is replicated, an observer cannot tell whether the *process* that produced the original was faithful [John et al., 2012, Fanelli, 2009].

What is missing, we argue, is not another call for transparency but a precise, checkable concept. We call that concept *unforgeability*, and we give it its broadest, most fundamental reading: a property of *evidence*, not of any particular record. Evidence is that which warrants belief in a claim, and evidence can be easier or harder to forge; unforgeability measures that difficulty, and it is therefore a matter of degree rather than an all-or-nothing property. Under this reading, the new paradigm is not a cryptographic technique but a *system of community review and knowledge-establishment*: a way of organizing the certification of scientific claims so that confidence in a claim can rest on the unforgeability of its evidence chain and on the soundness of the reasoning from evidence to claim, rather than on reproducibility alone. The model for this is mathematics. A theorem is not accepted because anyone has reproduced it—there is no rerun of a theorem—but because its proof is a chain of steps that any competent mathematician can check, linked by rules of inference that are sound. Mathematical confidence rests on the integrity of a checkable chain plus the soundness of reasoning, not on rerunning; the proposal here is that science can adopt that same structure for the dimension of its own claims that reproducibility, by construction, cannot reach.

Two points bound this claim. First, unforgeability is a property of the evidence, not of the world it describes: it shows that a body of evidence resists being counterfeited after the fact, not that the underlying events actually occurred. A holder of a valid signing key can always commit a chain of events that never happened; what the keyholder cannot do is rewrite that chain afterward without leaving a detectable break. Second, this narrower guarantee is still load-bearing, because the integrity failures that matter for process trustworthiness—post-hoc editing, backdating, silent deletion, the concealment of detected gaps—operate precisely by rewriting the record after the fact. Unforgeability thus narrows the ways a claim can be false, without pretending to establish the rest. It gives the broader goal of *process trustworthiness*—evidence about the conduct of research that can itself be trusted—a precise, falsifiable, and deliberately bounded core.

The distinction between the goal and the property matters. "Process trustworthiness" describes an aspiration that is easy to affirm and hard to evaluate; "unforgeability" names a property that a skeptic can actually check, because checking it reduces to re-deriving hashes and signatures. A design that aims at unforgeability commits itself to concrete mechanisms—an append-only hash chain, canonical serialization, a device signature, and an external timestamp—each of which can fail or be attacked, and each of which can therefore be stated and scrutinized. This is the sense in which the concept is scientifically productive rather than merely rhetorical.

Our contributions are:

1. a clear separation of *reproducibility* (what was claimed) from *unforgeable process evidence* (how the claim was arrived at), and an argument that the latter underwrites a new paradigm of integrity complementing reproducibility;

2. a characterization of *unforgeability* as a graded property of evidence, together with its epistemic role as *defeater reduction*—eliminative rather than confirmatory—and an argument, drawn from the analogy with mathematical proof, that a community can ground confidence in knowledge

on checkable, integrity-preserving evidence chains rather than on rerunning alone;

3. a positioning of this account against prior provenance, electronic lab-notebook, content-provenance, proof-of-process, and philosophy-of-science work (§3);

4. a proof-of-concept exemplar—an open Evidence Package v1 specification and a prototype reference implementation (Academic Integrity Recorder), released as open source—demonstrating that the property is achievable with ordinary, local, privacy-preserving tooling;

5. a candid treatment of scope, limits, and a governance path for the social system that the paradigm implies.

Non-claim. Unforgeable process evidence secures the *evidence*, not the world it describes. It shows that a committed record has not been altered, reordered, or backdated since it was produced by the attested key, modulo the stated cryptographic assumptions. It does *not* show that the underlying events actually occurred as recorded, nor certify identity, authorship, originality, correctness, or academic integrity, and it cannot prove that all research activity was captured. This boundary is designed into the recording format and its exported manifest, not merely asserted in prose.

2 The Reproducibility Crisis

2.1 What the empirical record shows

The crisis is not anecdotal. In a 2016 *Nature* survey of 1,576 researchers, **70%** reported having failed to reproduce another group's experiment, and more than half reported failing to reproduce *their own* [Baker, 2016]. Among large coordinated replications, the Open Science Collaboration reran 100 psychology studies and found only **36%** reproduced the original statistically significant result, with effect sizes roughly half as large [Open Science Collaboration, 2015]. In preclinical cancer biology, the Reproducibility Project reproduced only a minority of high-impact findings [Errington et al., 2021], consistent with earlier industry reports that roughly **11%** (6 of 53) of landmark preclinical studies could be reproduced internally [Begley and Ellis, 2012] and that a major pharmaceutical pipeline observed reproducibility in only about one fifth of projects [Prinz et al., 2011].

These numbers are not a verdict on any discipline's honesty; they are a verdict on the *visibility* of process. As Ioannidis [2005] argued two decades ago, the prevailing incentive structure predicts a large fraction of false positives even without malfeasance, and John et al. [2012] documented that questionable research practices (p-hacking, selective reporting, undisclosed flexibility) are common and often unconscious.

2.2 Why reproducibility alone is insufficient

Reproducibility operates on the *output* side of research. It asks whether, given shared inputs, the same outputs reappear. Three gaps follow.

First, **the reporting gap**. Reproducibility depends on what authors choose to share. Studies show extensive under-disclosure of experimental design, and that the absence of a disclosure norm is itself a cause of non-reproduction [Nosek et al., 2015]. What is never written down cannot be reproduced, and the decision of what to write down is exactly the process reproducibility cannot see.

Second, **the selectivity gap**. Even with full data, reproducibility cannot distinguish a result obtained by pre-specified analysis from one obtained by exploring many analyses and reporting the best [Ioannidis, 2005, Gelman and Loken, 2014]. Preregistration addresses this prospectively [Nosek et al., 2018], but it covers a narrow slice of ordinary research behavior and is silent about the long, messy trail of reading, drafting, and revision that precedes any registration.

Third, **the provenance gap**. Modern research increasingly involves tools—search, writing environments, version control, and AI assistants—whose use is rarely disclosed in a verifiable way [Gao et al., 2023]. Whether and how an AI contributed to a manuscript, whether a passage was copied from a collaborator, or whether a figure was regenerated after peer review are questions about *process*, and reproducibility is structurally blind to them.

None of this diminishes reproducibility; it motivates a second paradigm that observes the conduct of research directly rather than inferring it from its outputs.

3 Related Work

The idea that research conduct should be recorded and made verifiable has a long history, and the space between "result" and "process" is more crowded than the crisis framing alone suggests. Three strands are most relevant.

Scientific provenance. The W3C PROV family models provenance as a first-class, machine-checkable object [World Wide Web Consortium, 2013, Moreau and Missier, 2013, Moreau and Groth, 2013], and a substantial literature applies provenance to biomedical data and workflows; one 2024 scoping review alone synthesized 66 studies spanning provenance models, workflow tracking, integrity, reproducibility, and completeness [Gierend et al., 2024]. Electronic laboratory notebooks (ELNs) moved the paper notebook into software, and recent ELN systems such as IntegriLAB combine research-tool integration (GitHub, Overleaf, DataLad) with blockchain-anchored certification of activities and authorship [Easmin et al., 2026].

Content and media provenance. The C2PA standard provides cryptographically signed, tamper-evident manifests for digital media, binding origin, editing history, and AI involvement to content [Coalition for Content Provenance and Authenticity (C2PA), 2024]. It targets media artifacts rather than research conduct, but it establishes the pattern—a signed manifest verifiable without the producer's cooperation—that this paper applies to the research process itself.

Secure logging and proof-of-process. Tamper-evident append-only logs are well studied [Crosby and Wallach, 2009]. More recent "proof-of-process" work has made explicit the gap that motivates this paper: digital signatures prove key possession, not that the content was produced through a real process, and a signer can always reconstruct and sign intermediate states post hoc [Condrey, 2026].

Philosophy of science. The concepts this paper deploys are inherited, not coined. The notion of a *paradigm* as the organizing frame of a scientific community is Kuhn's [Kuhn, 1962]; the notion of a *defeater*—a consideration that undermines the warrant for a belief—is Pollock's, and the distinction between *rebutting* and *undercutting* defeaters is his [Pollock, 1986]. And the idea that experimental knowledge has always depended on making a result *witnessable* to a community of remote assessors—rather than on any single observer's testimony—is the central claim of the historical study of experimental life that runs from Robert Boyle's "virtual witnessing" onward [Shapin and Schaffer, 1985]. It is no accident that that study devotes a chapter to *replication and its troubles*: the difficulty of securing belief through rerunning, and the recourse to witnessable records when rerunning fails, is as old as experimental science itself.

What the present work adds is not provenance tracking itself—that is well established—but a *repositioning.* Existing systems largely aim to make the record *complete* or *certified*; they treat the record as the object of interest. This paper argues that unforgeability is a property of *evidence*—the graded difficulty of counterfeiting it—and that the contribution of a provenance system is best understood not as a guarantee of authenticity but as the elimination of a specified class of defeaters, explicitly bounded by what it does not establish. We develop that account in §4.

4 Unforgeability: Rationale and Epistemic Role

4.1 The warrant of a scientific claim

A scientific result functions as evidence for a claim about the world to the degree that the process producing it can be trusted. That trust decomposes into two questions that are easy to conflate but distinct in what they can be checked. The first asks whether the reported result follows from the reported materials—call this the question of *reproducibility*, of *what* was claimed. The second asks whether the reported process is the *actual* process—whether the experiment, the reading, the analysis, and the writing happened as they are described, in the order described, and before the outcome was known. Call this the question of *fidelity*, of *how* the claim was arrived at. A result can score high on the first question and low on the second, and vice versa.

Science has always treated the second question as load-bearing, even when it lacked the machinery to answer it mechanically. The laboratory notebook is a centuries-old technology whose entire purpose is to make the conduct of experiment visible to later scrutiny, precisely because its contemporaneity—the fact that its entries were written as events unfolded rather than reconstructed afterward—is what gives it evidentiary weight [Macrina, 2014]. The experimental tradition that runs from Robert Boyle onward secured the credibility of a "matter of fact" by making it *witnessable*: detailed reporting and "virtual witnessing" allowed distant readers to assess a result as if they had been present [Shapin and Schaffer, 1985]. The W3C PROV model formalizes provenance as a first-class, machine-checkable object [World Wide Web Consortium, 2013, Moreau and Missier, 2013, Moreau and Groth, 2013], and open-science movements have long insisted that transparency of process is as important as availability of products [Nosek et al., 2015, Fanelli, 2011]. What these strands share is an implicit commitment to exactly the property we wish to make explicit: that the account of how a result was produced should itself be trustworthy.

The reproducibility crisis can be read as the failure of the present ecosystem to secure this second question. Its signature behaviors—p-hacking, HARKing, selective reporting, the post-hoc construction of a clean narrative—do not primarily attack the regenerability of a result; they attack the fidelity of its provenance, and they survive reproduction precisely because a fabricated-but-consistent artifact can be re-derived just as easily as a genuine one. Reproducibility, by construction, cannot tell the two apart: it verifies the artifact, not the act. If the act is what matters for evidentiary weight, then a property that binds the act—rather than the artifact—is what is missing.

4.2 The mathematical analogy

Consider how confidence is established in mathematics. A theorem commands assent not because anyone has *reproduced* it—there is no experimental rerun of a theorem—but because its proof is a chain of steps, each checkable by any competent mathematician, linked by rules of inference that are sound. Two features carry the entire weight of mathematical confidence: the *integrity* of the proof chain (a single slipped step, a gap, or a substituted lemma breaks it), and the *soundness* of the reasoning that links step to step. If either fails, the theorem falls. Neither feature has anything to do with reproducibility: a proof is not more credible for having been re-derived twice, and a false theorem is not made true by many faithful re-derivations.

Mathematics thus demonstrates something the present crisis makes worth noticing. A community can ground its confidence in knowledge not on rerunning it, but on the checkability and integrity of the reasoning chain itself. Empirical science has, by contrast, concentrated its confidence mechanism almost exclusively on reproducibility— the empirical analogue of rerunning—and has left the corresponding chain, from evidence to conclusion, largely unsecured and unchecked. The proposal of this paper is that science can borrow the mathematical confidence-structure for exactly that neglected dimension: make the chain of process evidence as checkable and as resistant to retroactive alteration as a proof chain, so that confidence in a scientific claim can rest partly on the integrity of its evidence chain and the soundness of the inference from evidence to claim, and not solely on reproducibility.

One qualification keeps the analogy honest. Mathematics is deductive: a valid proof confers certainty. Science is inductive: evidence confers probability, never certainty. The analogy is therefore not an analogy of *certainty* but of *structure*: both communities can rest confidence on a checkable chain plus sound inference, rather than on rerunning alone. Unforgeable process evidence cannot make an empirical claim certain any more than reproducibility can; what it can do is give the inductive inference the same kind of checkable, integrity-preserving backbone that proof gives to deduction. This is the sense in which the mathematical analogy is a model for the paradigm, not a promise of mathematical certainty.

4.3 Unforgeability as a property of evidence

Unforgeability, we propose, is a property of *evidence*, not of any particular record, and it is graded rather than all-or-nothing. Evidence is that which warrants belief in a claim, and evidence can be easier or harder to counterfeit. Consider two photographs of the same scene. One is a yellowed, creased print offered by an elderly woman in a remote village; the other is a visually identical image supplied by an engineer with ready access to synthesis tools. As images they are indistinguishable; as evidence they are not equally credible, because the two bearers face very different costs in producing a convincing counterfeit. The village photograph is embedded in circumstances—physical aging, no fabrication infrastructure, no obvious means—that make a counterfeit expensive and improbable; the engineer's image can be altered cheaply. The first has the higher *degree of unforgeability*, and that is what underwrites its greater evidentiary weight.

This graded, pre-theoretic notion is what cryptographic recording makes explicit and, within a bounded domain, nearly maximal. A hash chain and a device signature add no new *content* to evidence; they raise its degree of unforgeability by attaching a checkable difficulty to any after-the-fact counterfeit. The degree is never absolute—it is relative to a threat model and to the resources an adversary can bring—but it is made *explicit and checkable*, where the village photograph's degree is merely implicit and circumstantial. This is the deeper point: the formation of evidence, and therefore of trustworthy knowledge, has always depended on this largely unspoken notion of how hard a thing is to forge. Process evidence makes that notion explicit.

It is equally important to state what unforgeability is *not*. Evidence can be asked to vouch for several things, and only some of them can be fixed by a mechanism. What a mechanism can secure is that a committed record has not been altered since it was produced, and—with external anchoring—when it was produced. What no mechanism can secure is that the recorded events correspond to reality, that capture was complete, or that the recorded process is the process that actually occurred; a signer can always commit a chain of events that never happened. This boundary is what keeps the property honest: unforgeability attaches to the evidence's resistance to retroactive fabrication, and it is strongest exactly where it refuses to over-reach.

A corollary follows. The epistemic contribution of unforgeable evidence is primarily *eliminative* rather than directly *confirmatory*. A validated record does not establish that the scientific claim is true, or even that the entire research process occurred exactly as represented. It rules out, or substantially weakens, a specific class of *defeaters*—that the committed record was silently altered, reordered, rewritten, or backdated after the fact [Pollock, 1986]. It removes grounds for doubt rather than supplying positive proof of honesty or truth. That is a real and measurable gain, but it is a gain in the strength of a *warrant*, not a shortcut to certainty.

4.4 The paradigm as a system of community review and knowledge-establishment

It is important to see unforgeability not as a property one attaches to a record, but as the organizing concept of a social-epistemic system. A paradigm, in the sense that matters here, is a way a community organizes the certification of knowledge [Kuhn, 1962]. The reproducibility paradigm organizes certification around the question “can the result be rebuilt?”; it has produced an entire apparatus—data and code sharing, preregistration, replication consortia, journal data policies. The paradigm proposed here organizes a complementary apparatus around the question “is the evidence for the result what it claims to be, and can that be checked?” Its machinery is a set of norms and practices: researchers produce process evidence whose unforgeability can be verified; peers, journals, and archives run an offline verifier on it; and confidence in a claim may then appeal to the unforgeability of its evidence chain and to the soundness of the reasoning from evidence to conclusion—rather than to reproducibility alone.

This is not a claim that such a system would replace reproducibility, nor that it would certify honesty. It is the claim that scientific certification and confidence need not be monopolized by reproducibility, and that mathematics shows how a checkable, integrity-preserving chain can serve as an alternative—and complementary—ground of assent. The paradigm’s center of gravity is therefore institutional and epistemic, not technological. The cryptography that raises unforgeability is a means; and, as with the blockchain schemes sometimes invoked for the same purpose, it is best understood as an illustrative heuristic rather than the point itself. The point is the system of review that such evidence makes possible.

4.5 Complementarity with reproducibility

Unforgeability is not a substitute for reproducibility; the two secure different questions, and both are needed. Two clarifications guard against over-reading the relationship. First, reproducibility is itself narrower than is often assumed: in the standard usage it means that a reported result can be recomputed from the same data, code, and procedures—it establishes recomputability, not truth, and not the soundness of the process that produced the data. Second, a verified evidence chain reduces a *specified class* of provenance doubts; it does not convert a failure to reproduce into a purely methodological verdict. A result can be reproducible and yet rest on a fabricated or selectively edited account, and a result can be unforgeably recorded and yet be wrong, dishonest, or built on events that were themselves fabricated before recording.

The relationship is cleanest as a two-by-two space (Table 1). A result that is reproducible but whose evidence is forgeable can be rebuilt yet may rest on a fabricated account. A result whose evidence is unforgeable but which fails to reproduce still faces open questions of method, honesty, and unrecorded activity—its evidence chain is intact, but its inference remains doubtful. A result that is both reproducible and unforgeably recorded enjoys the strongest available evidentiary basis short of independent corroboration of the underlying phenomenon. And a result that is neither commands the least.

Two observations follow. First, the properties are *independent*: knowing that a result reproduces tells us nothing about whether its evidence was genuine, and knowing that its evidence was unforgeably recorded tells us nothing about whether the result will reproduce. This independence is exactly why unforgeability contributes evidence that reproducibility cannot supply. Second, the value of unforgeability is asymmetric and incremental: it does not raise the ceiling of what science can claim, but it raises the floor below which evidence cannot easily fall. By making the retroactive rewriting of the record detectable, it converts a category of integrity failures from matters of contested testimony into matters of checkable fact.

	Reproducible	Not reproducible
Unforgeable evidence	result regenerates *and* the evidence chain is intact (strongest)	evidence intact but result does not regenerate (provenance doubts reduced; method, honesty, or unrecorded activity remain possible)
Forgeable evidence	result regenerates but the evidence may be fabricated or edited	neither regenerable nor attested (weakest)

Table 1: Unforgeability and reproducibility secure different questions and are independent. Each cell indicates what the two properties jointly establish.

4.6 Illustrative mechanisms

The property would be idle if it were not attainable with ordinary tooling, on ordinary machines, without imposing surveillance. The mechanisms are standard and illustrative rather than novel, and we mention them only to establish feasibility, not to add technical apparatus.

Local-first capture. The evidence can be collected on the researcher's own machine, with no server, no telemetry, and sensitive content never leaving the device in cleartext. This addresses the principal objection to process recording—that it is surveillance—and aligns with data-minimization norms [Voigt and Von dem Bussche, 2017]. Local-first is a design choice in service of the paradigm's adoption, not a headline property.

Cheap, standard cryptography. An append-only hash chain commits each event to its predecessor, so that editing, deleting, or reordering any event breaks the chain [Crosby and Wallach, 2009]. A device signature attests the recorder [Bernstein et al., 2012]; canonical serialization makes hashing independent of incidental formatting [Rundgren, 2020]; authenticated encryption seals sensitive content so that verification can proceed without exposing it [Bernstein, 2008]; and an optional external timestamp—the mechanism behind blockchain-anchored timestamping—can prove existence at or before a point in time without trusting a notary [OpenTimestamps, 2018]. None of this is exotic; all of it is battle-tested.

Offline, adversarial verification. Because the property is mathematical, a skeptic needs no special access to check it. The verifier is a small, auditable program that re-derives every hash and signature; its success, rather than any party's assertion, is what licenses the inference that the evidence is genuine. This is the same principle that makes signed commits and checksums trustworthy without a trusted third party, and it is what distinguishes unforgeable process evidence from a self-reported methods section, which remains forgeable however detailed it is.

5 A Feasibility Exemplar: a Prototype Reference Implementation

We now illustrate—not exhaust—the argument with a deliberately brief account of Academic Integrity Recorder, an open-source *prototype* reference implementation of the Evidence Package v1, released under the AGPL-3.0 license and open to community contribution. It is a proof-of-concept, not a finished system: the specification is the stable artifact, while the code is an evolving demonstration that the property is attainable, with deliberately partial platform support. We therefore describe only the mechanisms that carry the argument— *unforgeability*, *local-first capture*, *honest capability disclosure*, *privacy-by-default*, and *offline verification*—and leave implementation detail to the project itself.

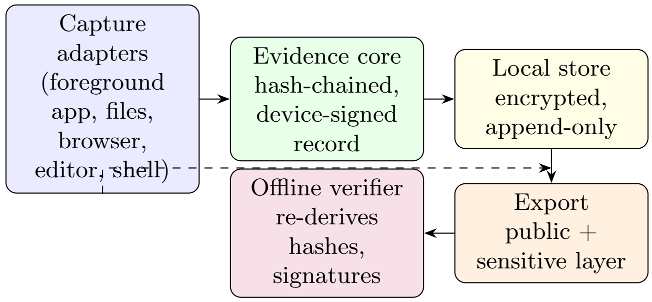

Figure 1: Sketch of the architecture. Evidence flows from capture adapters into an evidence core that maintains a hash-chained, device-signed record, is persisted in an append-only local store, exported as a two-layer Evidence Package v1, and verified entirely offline. No telemetry leaves the device.

5.1 Architecture

Capture adapters observe activity—foreground application, file events, browser and editor interactions, and (opt-in) shell commands. They forward events to the evidence core, which canonicalizes, hashes, signs, and persists them (Figure 1).

5.2 The hash chain

The core mechanism is an append-only hash chain. Each recorded moment is an *evidence event* carrying a sequence number, timestamps, a source, a semantic kind, and a sensitivity label; its payload hash and the hash of the preceding event are committed into the event's own hash. Because every link binds its predecessor, editing, deleting, or reordering any event changes its hash and so breaks every subsequent link and the signed head of the chain [Crosby and Wallach, 2009]. Events are persisted as immutable, append-only encrypted segments with signed checkpoints of the chain head; a crash is reconciled on open, and unsigned, missing, or reordered segments are rejected, so history cannot be silently rewritten.

5.3 Export and honest disclosure

Export produces a single Evidence Package v1 with a signed manifest and two layers. The *public layer* carries the event chain with payloads replaced by their hashes and a human-readable report, so integrity can be checked without revealing content; the *sensitive layer* holds the full events, artifacts, and disclosures, sealed under a passphrase-derived key. The manifest states exactly what the package claims—and, just as important, what it does not.

The design's distinctive mechanism is *honest capability disclosure*. Each capture capability is reported as Available, PermissionRequired, Degraded, or Unavailable, and any lapse in coverage is recorded as an explicit `Gap` event rather than silently omitted. A research claim can also be *anchored* to a specific passage of a manuscript and later revalidated, and an *active-time* measure reports a conservative lower bound on observed foreground activity. This honesty is what makes the record trustworthy: it is explicit about where it stops.

5.4 Offline verification

Because the property is mathematical, a skeptic can check it without special access: a small, auditable verifier re-derives every hash and signature from the package alone, and its success—not any party's assertion—licenses the inference that the evidence is genuine. Success proves *the integrity of the evidence chain and a device signature only*, and the verifier's own output says so.

6 Scope, Limitations, Governance, and Adoption

6.1 Scope and applicability

Unforgeable process evidence is most valuable where the contested question is *how*, not *whether*: disclosure of AI assistance, provenance of a passage, the trail of revision, the existence of a planned experiment that was later dropped, or a contemporaneous lab record. It strengthens theses, grant reports, and post-publication audit alike.

6.2 Limits of the property

Unforgeability is powerful but bounded, and the bounds belong to the property itself, not to any particular implementation.

- **Computational and conditional.** The property holds only while signing keys remain secret and any external timestamp is honest; a compromised key or a controlled clock can retrospectively undermine the record. Unforgeability is a guarantee under assumptions, not an absolute.
- **Attests the evidence, not the world.** An unforgeable record shows that the committed record has not been altered after commitment. It does not establish that the recorded events correspond to real occurrences, that the result is true, the method correct, or the work original; a faithfully recorded but flawed experiment remains flawed.
- **No completeness.** The property secures what was *recorded*, not that *everything* was recorded. Activity outside the instrumented environment is invisible to it, and no design can convert an absence of evidence into evidence of absence.
- **Not identity or authorship.** A device signature attests a key, not a legal person; it cannot distinguish the key's owner from anyone else using the device or account.
- **Cannot close the environment.** A determined adversary may work on uninstrumented machines or fabricate inputs before they are recorded; the property narrows the ways a claim can be false but does not eliminate every one.
- **Complements, not substitutes, review.** The record is evidence to be weighed alongside reproducibility and peer review, never a verdict on merit.

These limits are features when stated plainly: unforgeable evidence is trustworthy *because* it is explicit about where it stops.

6.3 Governance and adoption path

The paradigm is a system, and a system is built by norms and institutions, not by a technology alone. We therefore propose it as a lightweight, open community standard, in the spirit of how tamper-evident logging and code signing became default engineering expectations [Crosby and Wallach, 2009]. Concrete steps toward that system:

1. **Open specification and interop.** The format and its verification are specified openly, so any archive, journal, or reviewer can check a deposited package without vendor lock-in. Interoperable verification is what turns a proprietary tool into a community standard.

2. **Complement, don't replace.** Tie the record to existing norms: the data-availability statement, the preregistration, and the conflicts/disclosure form. The AI-disclosure entity maps directly onto emerging AI-use policies [Committee on Publication Ethics (COPE), 2023].

3. **Tiered adoption.** Start with self-archiving and reviewer-on-request, then optional journal encouragement, then (where appropriate) a deposition requirement for certain claim types. Voluntary, low-friction adoption avoids the surveillance objection.

4. **Incentives for honesty.** Because gaps are recorded, a complete, gap-light record becomes a positive signal; a record full of disclosed gaps is at least honest, which is more than the current default of silence. Over time, the community can learn to read these records as it reads a proof: not as a certificate of truth, but as a checkable account of how a result was reached.

7 Conclusion

Reproducibility verifies *what* was claimed; unforgeable process evidence verifies *how* the claim was arrived at—but only in a deliberately bounded sense. We have argued that this second axis underwrites a new paradigm of scientific integrity—not a replacement for reproducibility but a partner to it, closing the gap that reproducibility, by construction, cannot see. The paradigm is a system, not a tool: a way of organizing community review and the certification of knowledge so that confidence in a scientific claim can rest on the unforgeability of its evidence chain and the soundness of the reasoning from evidence to claim, rather than on reproducibility alone. Its model is mathematics, where confidence in a theorem has always rested on the checkability of the proof chain and the soundness of inference, never on rerunning—and where the same honesty about limits applies, since a checkable chain attests the inference, not the truth of its premises. Unforgeability is a graded property of evidence, and its epistemic role is defeater reduction: it makes a specified class of retrospective manipulations—editing, backdating, silent deletion—independently detectable, while remaining explicit about truth, authorship, and completeness, which it does not establish. The property is attainable today with local, privacy-preserving, cryptographic recording and offline adversarial verification, as an open specification and a prototype implementation demonstrate. Adopting it will not end the reproducibility crisis, but it gives the scientific community a new, precisely bounded instrument for the part of integrity that reproducibility was never designed to observe.

Acknowledgments

The author thanks the open-source and open-science communities whose tools and norms this work builds upon. The prototype reference implementation is released under the AGPL-3.0 license and welcomes community contribution.

References

Monya Baker. 1,500 scientists lift the lid on reproducibility. *Nature*, 533(7604):452–454, 2016. doi: 10.1038/533452a.

C. Glenn Begley and Lee M. Ellis. Drug development: Raise standards for preclinical cancer research. *Nature*, 483(7391):531–533, 2012. doi: 10.1038/483531a.

Daniel J. Bernstein. ChaCha, a variant of Salsa20. eSTREAM portfolio submission, 2008. URL https://cr.yp.to/chacha/chacha-20080128.pdf.

Daniel J. Bernstein, Niels Duif, Tanja Lange, Peter Schwabe, and Bo-Yin Yang. High-speed high-security signatures. In *Cryptographic Hardware and Embedded Systems – CHES 2012*, volume 7428 of *Lecture Notes in Computer Science*, pages 124–142. Springer, 2012. doi: 10.1007/978-3-642-23951-9_9.

Coalition for Content Provenance and Authenticity (C2PA). C2PA specification. Technical specification, version 2.1, 2024. URL https://c2pa.org/specifications/specifications/2.1/specs/C2PA_Specification.html.

Committee on Publication Ethics (COPE). Authorship and AI tools. COPE position statement, 13 February 2023, 2023. URL https://publicationethics.org/cope-position-statements/ai-author.

David Condrey. Witnessd: Proof-of-process via adversarial collapse. arXiv preprint arXiv:2602.01663, 2026. URL https://arxiv.org/abs/2602.01663.

Scott A. Crosby and Dan S. Wallach. Efficient data structures for tamper-evident logging. In *Proceedings of the 18th USENIX Security Symposium*, pages 317–334. USENIX Association, 2009. URL https://www.usenix.org/conference/usenixsecurity09/technical-sessions/presentation/crosby.

David L. Donoho, Arian Maleki, Inam Ur Rahman, Morteza Shahram, and Victoria Stodden. Reproducible research in computational harmonic analysis. *Computing in Science & Engineering*, 11(1):8–18, 2009. doi: 10.1109/MCSE.2009.15.

Rubaida Easmin, Mattia Veronese, Paul Allen, Giovanna Nordio, and Federico Turkheimer. Integrilab: a blockchain-enabled electronic lab notebook for reproducible neuroimaging research. *Frontiers in Neuroinformatics*, 20:1753937, 2026. doi: 10.3389/fninf.2026.1753937.

Timothy M. Errington, Maya Mathur, Courtney K. Soderberg, et al. Investigating the replicability of preclinical cancer biology. *eLife*, 10:e67401, 2021. doi: 10.7554/eLife.67401.

Daniele Fanelli. How many scientists fabricate and falsify research? a systematic review and meta-analysis of survey data. *PLoS ONE*, 4(5):e5738, 2009. doi: 10.1371/journal.pone.0005738.

Daniele Fanelli. Negative results are disappearing from most disciplines and countries. *Scientometrics*, 90(3):891–904, 2011. doi: 10.1007/s11192-011-0494-7.

Catherine A. Gao, Fred E. Howard, Nikolaus S. Markov, et al. Comparing scientific abstracts generated by ChatGPT to original abstracts using blinded human reviewers. *JAMA Network Open*, 6(2):e230178, 2023. doi: 10.1001/jamanetworkopen.2023.0178.

Andrew Gelman and Eric Loken. The statistical crisis in science. *American Scientist*, 102(6):460–465, 2014. doi: 10.1511/2014.111.460. URL https://www.americanscientist.org/article/the-statistical-crisis-in-science.

Kerstin Gierend, Frank Krüger, Sascha Genehr, Francisca Hartmann, Fabian Siegel, Dagmar Waltemath, Thomas Ganslandt, and Atinkut Alamirrew Zeleke. Provenance information for biomedical data and workflows: Scoping review. *Journal of Medical Internet Research*, 26:e51297, 2024. doi: 10.2196/51297.

John P. A. Ioannidis. Why most published research findings are false. *PLoS Medicine*, 2(8):e124, 2005. doi: 10.1371/journal.pmed.0020124.

Leslie K. John, George Loewenstein, and Drazen Prelec. Measuring the prevalence of questionable research practices with incentives for truth telling. *Psychological Science*, 23(5):524–532, 2012. doi: 10.1177/0956797611430953.

Thomas S. Kuhn. *The Structure of Scientific Revolutions*. University of Chicago Press, Chicago, 1962.

Francis L. Macrina. *Scientific Integrity: Text and Cases in Responsible Conduct of Research*. ASM Press, Washington, DC, 4 edition, 2014.

Luc Moreau and Paul Groth. *Provenance: An Introduction to PROV*. Synthesis Lectures on the Semantic Web: Theory and Technology. Morgan & Claypool, 2013. doi: 10.2200/S00528ED1V01Y201308WEB007.

Luc Moreau and Paolo Missier. PROV-DM: The PROV data model. Recommendation, World Wide Web Consortium (W3C), 2013. URL https://www.w3.org/TR/prov-dm/.

Brian A. Nosek, George Alter, George C. Banks, et al. Promoting an open research culture. *Science*, 348(6242):1422–1425, 2015. doi: 10.1126/science.aab2374.

Brian A. Nosek, Charles R. Ebersole, Amanda C. DeHaven, and David T. Mellor. The preregistration revolution. *Proceedings of the National Academy of Sciences*, 115(11):2600–2606, 2018. doi: 10.1073/pnas.1708274114.

Open Science Collaboration. Estimating the reproducibility of psychological science. *Science*, 349 (6251):aac4716, 2015. doi: 10.1126/science.aac4716.

OpenTimestamps. Scalable, trust-minimized, distributed timestamping. Project documentation, 2018. URL https://opentimestamps.org/.

John L. Pollock. *Contemporary Theories of Knowledge*. Rowman & Littlefield, Totowa, NJ, 1986.

Florian Prinz, Thomas Schlange, and Khusru Asadullah. Believe it or not: how much can we rely on published data on potential drug targets? *Nature Reviews Drug Discovery*, 10(9):712, 2011. doi: 10.1038/nrd3439-c1.

Anders Rundgren. JSON canonicalization scheme (JCS). Request for Comments 8785, Internet Engineering Task Force (IETF), 2020. URL https://www.rfc-editor.org/rfc/rfc8785.

Steven Shapin and Simon Schaffer. *Leviathan and the Air-Pump: Hobbes, Boyle, and the Experimental Life*. Princeton University Press, Princeton, NJ, 1985.

Paul Voigt and Axel Von dem Bussche. *The EU General Data Protection Regulation (GDPR): A Practical Guide*. Springer, 2017.

World Wide Web Consortium. PROV model primer. Working group note, World Wide Web Consortium (W3C), 2013. URL https://www.w3.org/TR/prov-primer/.